

Chapter 3 — Indications and Interpretation of Laboratory Testing

- Serum, urinary and radiological studies complement examination; **knowing when to order each from the history and findings** identifies pathology.
- **Urinalysis is among the commonest laboratory studies**, sometimes performed in the office. Dipsticks persist in small clinics and as backup; **most urine now runs on semi-automated systems using the same reagents**.

URINALYSIS

- **Fundamental, non-invasive test** for almost all patients with urinary symptoms; clues to infection, metabolic derangement, hydration and wide-ranging disease.
- **A complete urinalysis has three parts:** gross examination → biochemical analysis → microscopic analysis of the urinary sediment.
- **Microscopy of the sediment is indicated** for: an abnormal dipstick (e.g., microscopic haematuria); suspected or known nephrolithiasis; acute kidney injury (AKI); the initial evaluation of chronic kidney disease (CKD).

Collection of Urinary Specimens

- Wash hands → cleanse genitalia → **midstream specimen** into a sterile, dry container.
- **Examine within 2 hours at room temperature**, else **refrigerate at 2–8 °C** and rewarm before examination. Delay causes **bacterial overgrowth, pH change and disintegration of red blood cell (RBC) and white blood cell (WBC) casts**; refrigeration causes **crystal precipitation that may not redissolve**.

Males

- **Midstream clean catch:** retract foreskin → cleanse glans and meatus → start voiding into toilet → collect midstream in a wide-mouth sterile container.

Females

- Routine: **separate labia → cleanse introitus and meatus → midstream**. When sterility matters, **consider a catheterised specimen**.

Neonates, Infants and Children

- Depends on **toilet training**: trained children give a clean voided specimen; otherwise clean-catch void, adhesive sterile bag, transurethral catheterisation or suprapubic aspiration (SPA).

PAEDIATRIC COLLECTION METHODS COMPARED

Method	Contamination	Practicalities	Use for culture
Suprapubic aspiration	Lowest	Less successful; more invasive and painful	Yes
Transurethral catheterisation	Intermediate	More successful than SPA; less invasive, less painful — often preferred	Yes
Sterile urine bag	Highest	Non-invasive	No — not routinely ; may have a role for biochemistry only

- For **biochemical analysis alone**, catheterisation and SPA remain preferred, but a **bag may avoid a more invasive collection**.

Concept — *Every collection decision trades contamination against invasiveness; the purpose of the sample (culture versus chemistry) sets how much contamination can be tolerated.*

Gross Examination of Urine

- Visual and physical examination: **colour, turbidity and odour**.

Colour

- Pale yellow from **urochrome**; **concentration is the commonest cause of variation**, then foods, drugs, metabolites, infection.
- **Bright red/pink** → possibly active bleeding; **purple/brown** → old haemorrhage or retained clot; **brown/cola** → glomerular disease.

TABLE 3.1 COMMON CAUSES OF ABNORMAL URINE COLOUR

Colour	Physiological / pathological	Foods	Drugs and agents
Colourless	Very dilute urine; overhydration	—	—
Cloudy / milky	Phosphaturia; pyuria; chyluria; funguria	—	—
Red	Haematuria; haemoglobinuria / myoglobinuria; chronic lead and mercury poisoning	Anthocyanin (beetroot, blackberries)	Phenolphthalein (bowel evacuants); phenothiazines (e.g., prochlorperazine [Compazine]); rifampicin
Orange	Dehydration	—	Phenazopyridine (Pyridium); sulfasalazine (Azulfidine)
Yellow	Normal	—	Phenacetin; riboflavin
Green-blue	Biliverdin; indicanuria (tryptophan indole metabolites)	—	Amitriptyline; indigo carmine; methylene blue; phenols (intravenous [IV] cimetidine, IV promethazine); resorcinol; triamterene
Brown	Urobilinogen; porphyria	Aloe, fava beans, rhubarb	Chloroquine, primaquine; furazolidone; metronidazole; nitrofurantoin
Brown-black	Alkaptonuria (homogentisic acid); haemorrhage; melanin; tyrosinosis (hydroxyphenylpyruvic acid)	—	Cascara, senna (laxatives); methocarbamol; methyldopa; sorbitol

From Hanno PM, Wein AJ: *A Clinical Manual of Urology*, 1987.

- **Purpose and trap:** red or dark urine is not synonymous with blood — pigments, foods and drugs mimic haematuria, so colour prompts microscopy; the same agents also disturb dipstick colour development.

Turbidity

- **Commonest cause of cloudy urine: phosphaturia** — benign phosphate precipitation in **alkaline** urine; intermittent, post-prandial, asymptomatic. **Pyuria** is the other common cause.
- With indwelling catheters or nephrostomy tubes, **smell and turbidity may influence the decision to treat** presumed infection.
- Rare causes: **chyluria** (abnormal lymphatic–urinary communication, lymph mixed with urine), lipiduria, hyperoxaluria, hyperuricosuria.

PHOSPHATURIA VERSUS PYURIA

Test	Phosphaturia	Pyuria
Add acetic acid	Immediate clearing	No clearing
Odour	Not offensive	Characteristic pungent odour
Dipstick / microscopy	Large amounts of amorphous phosphate crystals	Leucocyte esterase positive; leucocytes seen

Odour

- **Malodour does not reliably correlate with infection**, though ammonia from some urinary tract infections (UTIs) gives an acrid smell.
- **Sweet or fruity** → ketones; **maple syrup** → maple syrup urine disease; **musty or mousy** → phenylketonuria.

Concept — Gross examination only generates hypotheses: every abnormal colour, turbidity or odour has a benign mimic, so the visual finding is always confirmed chemically or microscopically.

BIOCHEMICAL EXAMINATION OF URINE (URINE DIPSTICK)

- Reagent pads give a **colorimetric change**; quick, cheap. **Dwell time and urine volume vary by manufacturer**. Semi-automated analysers **standardise timing and colour reading**, and **share every interference**.
- Pads: specific gravity, pH, blood, protein, glucose, ketones, urobilinogen, bilirubin, leucocyte esterase, nitrite.
- **Technique**: fully immerse pads in **fresh, uncentrifuged** urine → remove quickly, drawing the edge along the container rim to shed excess → **hold horizontally** for the reaction time → compare each pad with its colour chart. Excess urine or vertical holding lets reagents run between pads → faulty results.
- **Outdated or exposed strips** are a common cause of false results; **pad discoloration before immersion** gives them away.

DIPSTICK INTERFERENCES (CONSOLIDATED FROM THE CHAPTER)

Interferent	False negative	False positive / other
Ascorbic acid (vitamin C)	Glucose, bilirubin, blood (inhibits peroxidase reaction), leucocyte esterase	—
Phenazopyridine; other colouring agents	Leucocyte esterase	Nitrite, bilirubin (phenazopyridine); ketones
Highly buffered alkaline urine	Protein	Falsely low specific gravity
High specific gravity	Leucocyte esterase; reduced glucose sensitivity	Blood (first-voided urine); ketones (if also acidic)
Dilute urine	Protein	RBC lysis impairs RBC assessment
Semen; menstruation; dehydration; vigorous exercise	—	Blood
Glycosuria, proteinuria, urobilinogen	Leucocyte esterase	—
Levodopa metabolites, mesna, sulfhydryl compounds	—	Ketones
Specimen contamination	—	Leucocyte esterase and nitrite — the major cause of false positives for both

Specific Gravity and Osmolality

- Range **1.001–1.035**; tracks osmolality; reflects hydration, renal function, solute load and **concentrating ability**.
- **<1.008 dilute; >1.020 concentrated; fixed at about 1.010 in acute or chronic renal insufficiency**.

SPECIFIC GRAVITY: CAUSES OF DECREASE VERSUS INCREASE

Decreased	Increased
Diuretics; increased fluid intake; diabetes insipidus ; reduced renal concentrating ability	Glycosuria (uncontrolled diabetes mellitus); syndrome of inappropriate antidiuretic hormone secretion; dehydration; IV iodinated contrast and dextran (can exceed 1.035)

- Osmolality (dissolved particles) **50–1200 mOsm/L**, same influences; abnormal → test renal function.
- **Very dilute urine (below 308 mOsm) lyses RBCs**, precluding reliable assessment.

pH

- Physiological **4.5–8.0**, average 5.5–6.5; **methyl red + bromothymol blue** read pH 5–9. **<5.5 acidic; >6.5 alkaline**.
- **Urine pH mirrors serum pH** by renal compensation (acidosis → acidic urine); **renal tubular acidosis (RTA) is the exception** — acidic serum, alkaline urine.

RENAL TUBULAR ACIDOSIS: TYPE I VERSUS TYPE II

Feature	Type I	Type II
Defect	Impaired tubular hydrogen ion excretion	Impaired bicarbonate reabsorption
Urine in severe metabolic acidosis	Always alkaline	May become acidic

- **Diagnosis of RTA: inability to acidify urine below pH 5.5 after an acid load**; low urine ammonium with high urine pH suggests RTA; **risk of calcium phosphate stones**.
- Presumed UTI with **pH >7.5 → urea-splitting organism, commonly *Proteus***: urease splits urea → ammonia → ammonium, raising pH → precipitation of magnesium ammonium phosphate (struvite) with calcium phosphate → rapid growth → **staghorn calculi**.
- **Uric acid and cystine stones form in acidic urine**; urinary alkalisation is central to therapy, with frequent pH monitoring to confirm adequacy.

Concept — *Urine pH normally follows serum pH; its diagnostic value lies in **dissociation** — alkaline urine despite acidaemia (RTA) or alkaline urine generated by bacterial chemistry (urease).*

Blood/Haematuria

- Normal: **<3 erythrocytes per high-power field (HPF)**. A haem-positive pad means **haematuria, haemoglobinuria or myoglobinuria** — it **suggests but never confirms RBCs**.
- Chemistry: **peroxidase-like activity of haem** oxidises a chromogen, colour proportional to haemoglobin.
- **Free haemoglobin/myoglobin → uniform field colour change; intact RBCs lyse on the pad → speckled dots**; more intact cells, more dots. Detects **as few as 5 RBC/μL**; dots coalesce above about 250 RBC/μL.

RED URINE: HAEMATURIA VERSUS PIGMENTURIA

Condition	Centrifuged pellet	Supernatant	Dipstick of supernatant	Plasma
Haematuria	Red; many RBCs	Not red	—	—
Haemoglobinuria	No RBCs	Red	Haem positive	Red
Myoglobinuria	No RBCs	Red	Haem positive	Clear
Drugs, foods, dyes, porphyria	No RBCs	Red	Haem negative	—

- **Why plasma differs**: haemoglobin is large, **haptoglobin-bound** and relatively insoluble, so it is poorly filtered and appears in urine only once **haptoglobin is saturated** — plasma stays red. Myoglobin is **small, water-soluble and not haptoglobin-bound**, so it is filtered immediately — plasma stays clear.

- Against microscopy (microhaematuria ≥ 3 RBC/HPF), **dipstick sensitivity is high but specificity lower** — more false positives.
- False positives (table above): note exercise causes **real but transient, benign** haematuria. **False negatives are rare**; ascorbate-oxidising strips counter ascorbic acid.
- Screening efficacy is controversial. **Confirm a haem-positive dipstick by microscopy before imaging or endoscopy**; the American Urological Association (AUA) / Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction (SUFU) Microhematuria Guideline (2020) and others agree **a dipstick alone is inadequate to diagnose haematuria**.

Differential Diagnosis and Evaluation of Haematuria

- **First step: nephrological versus urological, glomerular versus non-glomerular** — this decides further tests. Nephrological haematuria: often casts, **almost always significant proteinuria**.
- **Urological bleeding, even gross, rarely raises urinary protein to 100–300 mg/dL (2+ to 3+)**; proteinuria of this magnitude almost always means glomerular or tubulointerstitial disease → **nephrology consultation**.
- Morphology: glomerular RBCs are **dysmorphic**; tubulointerstitial and urological RBCs are **uniformly round** — they may lose haemoglobin ("ghost cells") but the shape stays round.

GLOMERULAR VERSUS NON-GLOMERULAR HAEMATURIA

Feature	Glomerular	Non-glomerular — medical (renal)	Non-glomerular — surgical (urological)
RBC morphology	Dysmorphic	Circular	Circular
RBC casts	Present	Absent	Absent
Proteinuria	Significant	Frequently significant	Minimal, even with heavy bleeding
Colour clue	Brown / cola	—	—

Glomerular Haematuria

- Suggested by **dysmorphic RBCs, RBC casts, proteinuria and brown/cola urine**. However, **a minority of biopsy-proven glomerulonephritis (GN) shows haematuria alone**, without casts or proteinuria.

TABLE 3.2 GLOMERULAR DISORDERS IN PATIENTS WITH GLOMERULAR HEMATURIA

Rank	Disorder	Group
1	Immunoglobulin A (IgA) nephropathy (Berger disease) — about one-third of cases	Primary GN
2–3	Mesangioproliferative GN; focal segmental proliferative GN	Primary GN
4	Familial nephritis (e.g., Alport syndrome)	Hereditary
5–7	Membranous GN; mesangiocapillary GN; focal segmental sclerosis	Primary GN
Less common	Systemic lupus erythematosus (SLE); postinfectious GN; subacute bacterial endocarditis	Secondary / systemic
—	Unclassifiable; others	—

Modified from Fasset RG, Horgan BA, Mathew TH. *Lancet*, 1982.

- **Pattern:** IgA nephropathy leads by a wide margin; primary GNs dominate; hereditary nephritis ranks high (take a family history). **Trap:** systemic causes are dramatic but uncommon.

FIG. 3.1 EVALUATION OF GLOMERULAR HAEMATURIA — CLINICAL CLUE TO DIAGNOSIS

History	Key test / finding	Diagnosis
Family history of haematuria or abnormal urinalysis	+ deafness	Familial nephritis → Alport nephritis

History	Key test / finding	Diagnosis
Rash, arthritis	Low C3, C4; positive antinuclear antibody (ANA)	SLE
Haemoptysis, bleeding tendency	Microcytic anaemia	Goodpasture syndrome
Recent upper respiratory or skin infection / rash	Raised antistreptolysin O (ASO) titre, low C3	Poststreptococcal GN
Related to exercise	Normal ASO and C3 → renal biopsy for IgA, IgG, β 1c-globulin	IgA nephropathy
No other symptoms or signs	Serum creatinine, creatinine clearance, 24-hour urinary protein → renal biopsy	Mesangioproliferative, mesangiocapillary or membranous GN

- **Young male with low-grade fever and erythematous rash** → IgA nephropathy; **child after streptococcal throat or skin infection** → poststreptococcal GN.
- Creatinine, clearance and 24-hour protein/albumin **quantify dysfunction**; further tests separate **immune from non-immune** causes; **renal biopsy** (light, immunofluorescence, electron microscopy) when it will change treatment.

IgA nephropathy (Berger disease)

- **Commonest cause of glomerular haematuria.** Children and young adults, **male predominance**.
- **Haematuria after upper respiratory tract infection or exercise**; most have no systemic symptoms; gross haematuria intermittent, microscopic may be constant.
- Chronic; prognosis mostly excellent, but **a minority progress to renal insufficiency**.
- **Poor prognosis:** older age at onset, abnormal initial renal function, persistent proteinuria, hypertension.
- Pathology: focal or segmental **mesangial proliferation; IgA, IgG and β 1c-globulin deposits** (mesangial IgA/IgG also occurs in other GNs). Role of IgA uncertain; viral trigger suspected, unproven; exercise link unexplained.
- **Mimics:** Henoch-Schönlein purpura, SLE, bacterial endocarditis, Goodpasture syndrome.
- **RBC casts establish glomerular origin; without casts, a urological evaluation** excludes a urinary tract source and confirms bilateral renal origin.
- **Confirmed by renal biopsy.** Once established, **repeat haematuria evaluations are generally not indicated**.
- No curative treatment; management is supportive (renin-angiotensin blockade), and renal function remains stable in most.

Concept — *Glomerular bleeding announces itself by what crosses a damaged filter* — **dysmorphic cells, casts and protein** — *and each clinical clue points to the immune process causing the damage.*

Nonglomerular Haematuria

Medical/nonsurgical

- Renal-origin causes are **tubulointerstitial, renovascular or systemic**. Urinalysis: **circular RBCs, no RBC casts**, frequently **significant proteinuria** — which separates these nephrological diseases from urological disease.

NON-GLOMERULAR MEDICAL HAEMATURIA — HISTORY CLUE TO CAUSE

Clue	Consider	Next step
Family history of haematuria or bleeding tendency	Blood dyscrasia	Investigate further

Clue	Consider	Next step
Family history of urolithiasis + intermittent haematuria	Inherited stone disease	Serum and urine calcium, uric acid
Family history of renal cystic disease	Medullary sponge kidney; adult polycystic kidney disease	Imaging
Diabetes; Black patients (sickle cell disease or trait); heavy analgesic use	Papillary necrosis	—
Supratherapeutic anticoagulation	Drug-induced haematuria	Full work-up nonetheless
Vigorous or long-distance exercise	Exercise-induced haematuria	Repeat after rest
Severe hypertension, flank/abdominal bruit, atrial fibrillation	Renal artery embolism/thrombosis; arteriovenous fistula; renal vein thrombosis	Renal vascular evaluation
Pelvic congestion	Nutcracker syndrome	—

- **Therapeutic anticoagulation does not predispose to haematuria**; urological disease is found as often as in controls. **Never attribute haematuria to the anticoagulant — full work-up.**
- **Exercise-induced**: runners (>10 km); at end of run, **clears rapidly with rest**; renal (dysmorphic cells in some) or bladder (punctate haemorrhagic lesions at cystoscopy); **may be the first sign of IgA nephropathy.**

Surgical

- Collecting-system pathology: **urological tumours, stones, benign prostatic hyperplasia (BPH), UTIs.**
- Circular RBCs, no casts; "**essential**" haematuria suggested by **absent significant proteinuria** — though renal disease may also lack proteinuria.
- **Asymptomatic microscopic haematuria (AMH) = ≥ 3 RBC/HPF on a properly collected specimen in the absence of an obvious benign cause** (AUA/SUFU 2020). The guideline covers diagnosis, initial evaluation, risk stratification, follow-up and an algorithm.

Concept — *Once glomerular disease is excluded, proteinuria is the discriminator between renal-parenchymal and collecting-system bleeding: heavy bleeding from the urinary tract barely raises urinary protein.*

Proteinuria

- Normal **80–150 mg/day**; concentration seldom **>20 mg/dL** (dilute urine can hide significant proteinuria below this).
- First sign of renovascular, glomerular or tubulointerstitial disease, or **overflow (multiple myeloma)**; also non-renal and physiological (strenuous exercise).
- Normal mix: **~30% albumin, 30% globulins, 40% tissue proteins (chiefly Tamm–Horsfall)**; electrophoresis of the shifted profile points to the cause.

Pathophysiology**GLOMERULAR, TUBULAR AND OVERFLOW PROTEINURIA**

Type	Mechanism	24-hour quantity	Main protein	Causes
Glomerular (commonest)	Increased glomerular capillary permeability	Suspect >1 g; almost certain >3 g; algorithm: >2 g, albumin-predominant, diagnostic	Albumin ≈ 70%	Primary GN (e.g., IgA); diabetes mellitus (commonest without haematuria); amyloidosis; arteriolar nephrosclerosis
Tubular	Failure to reabsorb normally filtered low-molecular-weight proteins	Seldom >2–3 g; algorithm 300–2000 mg	Low-molecular-weight globulins; albumin only 10–20%	Fanconi syndrome (glycosuria, aminoaciduria, phosphaturia, uricosuria); drug / heavy-metal intoxication; Balkan nephropathy; sarcoidosis
Overflow	No renal disease: raised plasma abnormal proteins exceed tubular reabsorption	300–2000 mg (algorithm)	Abnormal proteins — immunoglobulin light chains (Bence Jones)	Multiple myeloma (commonest); haemoglobinuria; myoglobinuria

- **Two glomerular thresholds coexist in the chapter:** >1 g (suspicion) and >3 g (near-certainty) index the likelihood of glomerular disease; >2 g with albumin predominance is the operational cut-off in the evaluation algorithm (Fig. 3.2). They answer different questions and are not in conflict.

Detection**DIPSTICK VERSUS SULFOSALICYLIC ACID**

Feature	Dipstick	3% sulfosalicylic acid precipitation
Principle	Tetrabromophenol blue ; pH-dependent colour shift, yellow → shades of green	Strong-acid precipitation of protein
Detection limit	20–30 mg/dL	15 mg/dL
Protein detected	Mainly albumin	Albumin and other proteins
False negatives	Alkaline urine; dilute urine; non-albumin protein	—

- Heavy proteinuria (>1 g/24 h) is seldom missed. **Dipstick negative, sulfosalicylic acid strongly positive → suspect myeloma** → Bence Jones testing.
- Positive → **quantify by 24-hour collection** → characterise: **electrophoresis separates glomerular from tubular; immunoassay and electrophoresis/immunofixation are methods of choice for Bence Jones protein.**

Evaluation

PROTEINURIA BY TIMING

Timing	Features	Action
Transient	Common, especially in children; resolves within days; fever, exercise, emotional stress; congestive heart failure in older patients	Repeat urinalysis; non-renal cause + negative repeat → no further evaluation ; persistent → evaluate
Intermittent — orthostatic	Young males; upright position only ; seldom >1 g/day; normal when recumbent; attributed to raised renal vein pressure on standing; often resolves spontaneously; no morbidity	Normal renal function → no further evaluation ; not position-related → evaluate as persistent
Persistent	Mostly glomerular	24-hour quantitation + qualitative protein analysis

Fig. 3.2 Evaluation of proteinuria (persistent arm): 24-hour protein + qualitative evaluation →

- >2000 mg, mainly albumin → **glomerular** → with dysmorphic RBCs and casts → Fig. 3.1 pathway; with little haematuria → **diabetes mellitus (commonest)** and other causes in the table.
- 300–2000 mg, mainly globulins → **immunofixation** → normal proteins = **tubular**; abnormal = **overflow** (Bence Jones = myeloma; haemoglobin; myoglobin).

Concept — Proteinuria is classified in two passes: timing decides whether to investigate at all, then quantity and composition identify the leak — glomerular filter, tubular reabsorption or plasma overload.

Glucose and Ketones

- Diabetes screen. Filtered glucose is reabsorbed proximally up to the **renal threshold ≈ serum glucose 180 mg/dL**.
- Chemistry (double sequential reaction): glucose + **glucose oxidase** → gluconic acid + hydrogen peroxide → peroxidase oxidises chromogen → colour. **Specific for glucose**; no cross-reaction with other sugars.
- Sensitivity falls with **rising specific gravity and temperature**.
- **Drug-induced glycosuria: sodium–glucose cotransporter 2 (SGLT2) inhibitors (gliflozins)**.
- Ketones appear when carbohydrate reserves are depleted and fat is broken down: **diabetic ketoacidosis, pregnancy, starvation, rapid weight loss**. Acetoacetic acid, acetone and β-hydroxybutyric acid are excreted; **ketones appear in urine before blood**.
- Dipstick: **sodium nitroprusside + acetoacetic acid → purple**; detects acetoacetate at 5–10 mg/dL but **not acetone or β-hydroxybutyrate**. Laboratory analysers quantify β-hydroxybutyrate down to 0.05 mmol/L.
- **Glucose-positive dipstick → test ketones**; diabetes suggested.

Bilirubin and Urobilinogen

- Normal urine: **no bilirubin, small amounts of urobilinogen**.

CONJUGATED VERSUS UNCONJUGATED BILIRUBIN

Feature	Conjugated (direct)	Unconjugated (indirect)
Origin	Hepatocyte — conjugated with glucuronic acid	Haem metabolism
Solubility / size	Water-soluble; low molecular weight	Water-insoluble; albumin-bound, higher molecular weight
In urine	Only with intrinsic hepatic disease or bile duct obstruction	Never , even in disease
Diazo reaction	Reacts without accelerant ("direct")	Needs accelerant

- **Urobilinogen = end product of bilirubin metabolism:** gut bacteria convert conjugated bilirubin; about half excreted in stool, half reabsorbed enterohepatically; **1–4 mg/day escapes to urine.**
- **Raised urinary urobilinogen:** haemolysis, hepatocellular disease. **Reduced:** bile duct obstruction or antibiotics altering gut flora (serum conjugated bilirubin rises).
- Dipstick: **diazonium salt** reaction; phenazopyridine mimics it (turns red in acid).

Concept — *The dipstick is a sensitive screen with known chemical blind spots: every positive that would trigger investigation needs microscopic or quantitative confirmation, and every negative is read against the interferents present.*

LEUKOCYTE ESTERASE AND NITRITE TESTS

- **Leucocyte esterase → WBCs; nitrite → bacteriuria;** both are used to screen for UTI.
- **Most accurate diagnosis of infection: sediment microscopy for pyuria + urine culture.**
- A "directed" urinalysis that reflexes to culture when abnormal reduces unnecessary cultures and avoids missing specimens that need one.
- **Leucocyte esterase positive, nitrite negative → consider non-infectious inflammation;** obtain microscopy and culture before any empirical antibiotic.

LEUKOCYTE ESTERASE VERSUS NITRITE

Feature	Leucocyte esterase	Nitrite
Detects	Host response — WBCs	Organism — bacteriuria
Chemistry	Esterase from lysed neutrophils and macrophages hydrolyses indoxyl carbonic acid ester → indoxyl → oxidises diazonium chromogen	Nitrate reductase (gram-negatives, especially <i>Enterobacteriaceae</i>) converts nitrate → nitrite → diazotisation → red azo dye
Reading	At 5 minutes; sensitivity then falls as leucocytes lyse	—
Performance	Negative when bacteriuria lacks significant pyuria	High specificity, low and variable sensitivity; less accurate below 10 ⁵ organisms/mL
Characteristic miss	Glycosuria, proteinuria, certain infections	Enterococci (few have nitrate reductase)
Commonest false positive	Contamination	Contamination

- **Leucocyte esterase screening must always be paired with nitrite testing.**

Fig. 3.3 Protocol for deciding the need for sediment microscopy in an asymptomatic population: urine sample → assess visual appearance → **cloudy → send to laboratory;** clear → dipstick for nitrite and

leucocytes → **nitrite positive and/or leucocytes +, ++ or +++ → send to laboratory**; nitrite negative and leucocytes negative or trace → discard. (From Flanagan PG et al. *Lancet*, 1989.)

- This older protocol detected most infected specimens and cut microscopy, but **other studies found the dipstick no replacement for microscopy**. Automated urinalysis, reflex culture and requisition quality-improvement now reduce unnecessary tests.
- **A dipstick positive for only one of leucocyte esterase or nitrite → confirm with microscopy and culture before antibiotics. Dipstick testing has not been shown to replace sediment microscopy.**

Concept — *Leucocyte esterase and nitrite are complementary halves: esterase reads the host response (sensitive, non-specific), nitrite reads the organism's chemistry (specific, insensitive) — neither, nor both together, replaces microscopy and culture.*

Urinary Sediment

Obtaining and Preparing the Specimen

- Clean-catch midstream; **harder in women (introital leucocytes and bacteria)**.
- **Catheterised sample for culture** when findings are equivocal for true UTI, when contamination is expected or hard to avoid, or when concerning results are **at odds with a benign clinical picture**.
- **First-voided specimen is the specimen of choice; examine within 1 hour.**

Preparation: centrifuge **10–15 mL for 5 minutes at 3000 rpm** → pour off supernatant → resuspend sediment in **0.3 mL saline** by tapping the tube → **pipette** the residual fluid onto the slide (pouring deposits excess) → ideal volume **0.01–0.02 mL** → lower the coverslip **edge first** so fluid rises by capillary action and air is expelled. Dropping the coverslip flat disperses urine and traps bubbles that distort examination.

Microscopy Technique

- **Low (×100) and high (×400) power; ≥10–20 fields**; vary fine focus; **oil immersion seldom needed**. Scan the whole coverslip at low power, especially **the edges, where casts concentrate**.

LOW POWER VERSUS HIGH POWER

Low power (×100) — sufficient for	High power (×400) — necessary for
Erythrocytes, leucocytes, casts, cystine crystals , oval fat macrophages, parasites (<i>Trichomonas vaginalis</i> , <i>Schistosoma haematobium</i>)	Circular versus dysmorphic erythrocytes ; other crystal types; bacteria and yeast

- Polarised light and phase-contrast microscopy help define formed elements and crystals.
- Automated analyser microscopy saves time and cost, but **manual microscopy performs better in pathological states such as AKI**.
- **Examine for six elements: cells, casts, crystals, bacteria, yeast, parasites.**

Cells

- RBC morphology at high power; **routine brightfield usually suffices. Condenser at lowest aperture** dims background → phase-like effect, membranes stand out.

CIRCULAR VERSUS DYSMORPHIC ERYTHROCYTES

Circular (non-glomerular)	Dysmorphic (glomerular)
Even haemoglobin distribution; round, crenated or biconcave-disc contour	Irregular shape ; minimal haemoglobin; irregular cytoplasmic distribution

- Automated erythrocyte typing (e.g., Coulter counter) is accurate but **not accepted into routine urological practice and probably unnecessary**.
- **Erythrocyte mimics: yeast (distinguished by budding) and fat droplets (highly refractile).**

- Leucocytes: identified at low power, confirmed at high power. **Neutrophils commonest** (granular cytoplasm, multilobed nuclei); **eosinophils raise concern for acute interstitial nephritis**.
- Normal: 1–2 WBC/HPF in men; up to 5/HPF in women** (vaginal contamination); more indicates infection or inflammation.

OLD VERSUS FRESH LEUCOCYTES

Old leucocytes	Fresh leucocytes
Small, wrinkled; common in vaginal secretions of normal women	Larger, rounder; indicate urinary tract pathology ; at specific gravity <1.019 cytoplasmic granules show glitter-like movement — " glitter cells "

EPITHELIAL CELLS COMPARED

Feature	Squamous	Transitional	Renal tubular
Origin	Lower urethra, trigone of postpubertal females, vagina; frequent in female specimens	Remainder of the urinary tract	Renal tubules
Appearance	Large ; small central nucleus about the size of an erythrocyte; irregular, finely granular cytoplasm	Smaller than squamous; larger nucleus; prominent perinuclear granules	Slightly larger than leucocytes; hard to distinguish from them
Significance	Contamination	Malignant cells : altered nuclear size and morphology — Papanicolaou staining or automated flow cytometry	Least common, most significant: always indicates renal pathology

Casts

- Cast = **cylindrical protein coagulum formed in the tubular lumen**, trapping luminal contents. Favoured by **low urine flow, high concentration, low pH**.
- Tamm-Horsfall mucoprotein is the basic matrix of all casts**; secreted by tubular epithelium and always present in urine. Casts are named by their contents.

CASTS AND THEIR SIGNIFICANCE

Cast	Significance
Hyaline (mucoprotein only)	May be non-pathological — after exercise or heat exposure; also pyelonephritis, chronic renal disease
RBC	Diagnostic of glomerular bleeding — usually GN; sometimes acute interstitial nephritis
WBC	Acute GN; acute pyelonephritis; acute tubulointerstitial nephritis
Cellular (sloughed renal tubular epithelial cells)	Non-specific renal damage
Granular, waxy	Further degeneration of cellular elements
Haem-granular ("muddy brown")	Acute tubular necrosis
Fatty	Nephrotic syndrome; lipiduria; hypothyroidism

Crystals

- Particularly useful in stone disease to suggest aetiology; **crystals occur in both normal and pathological states**.
- Cystine crystals establish the diagnosis of cystinuria** — the one crystal that is never normal.

CRYSTALS BY URINE PH (WITH FIG. 3.16 SHAPES)

Urine pH	Crystals	Appearance (Fig. 3.16)
Acidic	Uric acid; cystine	Uric acid: rhomboid / lemon-shaped plates; cystine: hexagonal plates
Alkaline	Calcium phosphate; triple phosphate (struvite)	Struvite: coffin-lid prisms
Acidic or alkaline	Calcium oxalate	Envelope (octahedral) shape
Unrelated to pH	Cholesterol — rare; in lipiduria, remains in droplet form	—

Bacteria

- Normally absent (**asymptomatic bacteriuria rises with age**); in a fresh uncontaminated specimen, suggest UTI.
- Each HPF views 1/20,000–1/50,000 mL, so **1 bacterium/HPF $\approx$ >30,000/mL** and **5 bacteria/HPF $\approx$ 100,000 (10^5)/mL** — the standard UTI threshold in a clean-catch specimen.
- That threshold applies mainly to **women** (frequent contamination); **any bacteria in a properly collected male midstream specimen $\rightarrow$ culture**.
- High-power morphology: **gram-negative rods — bacillary; streptococci — beaded chains; staphylococci — clumps**.

Yeast

- ***Candida albicans* is commonest**; oval biconcave shape mimics erythrocytes and calcium oxalate crystals — **distinguished by budding and hyphae**.
- Seen in **diabetes mellitus, chronic illness, indwelling catheters**, or as contaminants in vaginal candidiasis.

Parasites**URINARY PARASITES**

Feature	<i>Trichomonas vaginalis</i>	<i>Schistosoma haematobium</i>
Disease	Frequent cause of vaginitis; occasionally urethritis in men	Urinary tract pathogen
Geography	—	Not in the United States; common in the Middle East and North Africa
Identification	Low power, clean-catch; large cells with rapidly moving flagella	Ova with a terminal spike

Expressed Prostatic Secretions

- Not strictly sediment, but **examine in any man with suspected prostatitis**.
- Normal fluid: few if any leucocytes; **increased or clumped leucocytes $\rightarrow$ prostatitis; oval fat macrophages $\rightarrow$ post-infection prostatic fluid**.
- Normal secretory granules resemble leucocytes but **lack nuclei**.

Figure legends

- **Circular RBCs** (Figs. 3.4–3.6): epithelial origin, bladder tumour, interstitial cystitis. **Dysmorphic RBCs** (3.7–3.8): Berger disease; granulomatosis with polyangiitis (Wegener) — dense deposits around the membrane.

- Old leucocytes with *Proteus* staghorn (3.10); RBC cast with distinct hyaline border (3.13); *Escherichia coli* (3.17); oval fat macrophage with **doubly refractile fat** (3.22).

Concept — Every sediment element is read for its **site of origin**: casts and tubular cells place the lesion in the kidney, dysmorphic cells in the glomerulus, squamous cells signal contamination — microscopy localises where the dipstick cannot.

KEY POINTS

- A complete urinalysis includes gross examination, dipstick chemical analysis and microscopic analysis of the urinary sediment.
- A dipstick alone is inadequate for the diagnosis of microscopic haematuria.
- AMH is defined as ≥ 3 erythrocytes/HPF on a properly collected specimen in the absence of an obvious benign cause.
- Haematuria should be stratified into glomerular, non-glomerular, medical and surgical causes.
- A dipstick positive for only leucocyte esterase or nitrite, but not both, should be confirmed with microscopy and urine culture.
- A catheterised specimen should be considered when uncontaminated voided specimens are difficult to obtain.

Serum Laboratory Studies

- Electrolytes, creatinine and estimated glomerular filtration rate (GFR) define renal function in obstruction and medical renal disease.
- **Urgency of intervention for obstruction is determined by renal function** (and subsequent imaging).
Check electrolytes in postobstructive diuresis.
- Presence or history of renal disease informs evaluation of haematuria, proteinuria and anatomical pathology.
- Serum potassium, carbon dioxide (bicarbonate) and liver transaminases assist evaluation of malignancy or multi-system metabolic derangement.
- **Albumin**: non-specific marker of debility and **postoperative recoverability**; as an acute-phase reactant it is **artificially low early in infection or systemic inflammation**.
- Complete blood count (CBC: RBC, WBC $\pm$ differential, haemoglobin, haematocrit, platelets): perioperative, malignancy follow-up, infection work-up; **haemoglobin assesses anaemia in ongoing haematuria**.

TUMOR MARKERS

Prostate-Specific Antigen

- Important, controversial; **interpret with digital rectal examination**.
- Prostate-specific antigen (PSA) = **human kallikrein peptidase 3 (hK3)**, a **serine protease** of the kallikrein family, produced by **prostatic luminal epithelial cells**; **highly specific for the prostate gland** (organ, not cancer).
- Sensitive for prostatic pathology → useful in BPH and prostatitis; **correlates with prostate volume** and prostatic inflammation.
- **One of the most reliable tests for following prostate cancer after treatment** — radiation, radical prostatectomy, hormonal therapy or immunotherapy.

Testis Tumor Markers

- α -fetoprotein (AFP), beta-human chorionic gonadotropin (β hCG) and lactate dehydrogenase (LDH), ordered for a scrotal or testicular mass found on examination or ultrasonography.
- **Measure at diagnosis → after radical orchidectomy → during chemotherapy (response) → after treatment (relapse).**

TESTIS TUMOUR MARKERS

Marker	Half-life	Raised in	Never raised in	Non-testicular causes
AFP	5–7 days	Embryonal; yolk sac	Pure seminoma; pure choriocarcinoma	Liver disease (cirrhosis, hepatitis); liver, pancreatic, gastric cancer; infants <1 year (physiological)
β hCG	24–36 hours	Embryonal; choriocarcinoma; seminoma	—	Marijuana; hyperthyroidism; hypergonadotropic hypogonadism (luteinising hormone cross-reacts) ; liver, pancreatic, gastric cancer
LDH	24 hours	Most commonly elevated , but non-specific	—	Kidney, liver and heart disease; other cancers; anaemia; infections. Isolated LDH rise usually not a reason to treat

Concept — Each marker is read against its **tissue of origin**: AFP maps to yolk-sac and embryonal elements and is therefore never made by pure seminoma or choriocarcinoma; β hCG maps to trophoblastic and seminomatous elements; LDH is abundant but histologically non-specific.

Urinary Markers in Bladder Cancer

- Cytology, formerly standard screening for urothelial cell carcinoma (UCC), is **no longer recommended for screening or initial AMH evaluation** (AUA/SUFU 2020).
- **Uses**: surveillance of known UCC; **persistent microhaematuria after a negative initial evaluation**; suspicious radiological abnormalities; suspected high-grade malignancy.
- **Highly sensitive for high-grade UCC; sensitivity falls markedly for low-grade**; highly specific for high-grade disease.
- Nuclear matrix protein 22 (NMP22), bladder tumour antigen (BTA) stat, ImmunoCyt and UroVysion fluorescence in situ hybridisation (FISH) are available but **not recommended for routine use or screening**; used in selected cases, generally not at initial evaluation.

Nephrolithiasis

- AUA medical management of kidney stones guideline (2014; validity confirmed 2019): **every newly diagnosed renal or ureteric stone patient → screening evaluation**: medical and dietary history + serum chemistries (basic metabolic panel, calcium; possibly intact parathyroid hormone [PTH] and uric acid) + urinalysis; **stone analysis at least once**.
- **High-risk or interested first-time stone formers → 24-hour urine** on usual diet: at least volume, pH, calcium, oxalate, uric acid, citrate, sodium, potassium, creatinine; $\pm$ cystine screening, magnesium, sulfate, urea nitrogen.
- **Fast and calcium-load testing are no longer routine** — subtyping hypercalciuria does not change treatment.

Adrenal Disorders and Lesions

- Cushing syndrome, Conn syndrome and pheochromocytoma meet urology via rising incidentalomas. **Every adrenal incidentaloma is tested for function: cortisol and catecholamine excess in all; aldosterone in hypertensive patients.**

ADRENAL FUNCTIONAL TESTING

Condition	Tests	Tested in
Hypercortisolism	Overnight low-dose dexamethasone suppression; late-night salivary cortisol; 24-hour urinary cortisol	All
Catecholamine excess	Plasma free fractionated metanephrines; 24-hour urinary fractionated metanephrines and catecholamines	All
Primary aldosteronism	Serum aldosterone >15 ng/dL + morning aldosterone-to-renin ratio >20 → adrenal vein sampling to lateralise	Hypertensive patients

Infertility, Hypogonadism, and Erectile Dysfunction

- Suspected hypogonadism: total and free testosterone, luteinising hormone, follicle-stimulating hormone, prolactin, thyroid-stimulating hormone, oestradiol.
- **Semen analysis is central** to assessing male fertility concerns.

KEY POINTS

- Serum creatinine and GFR should be ordered when renal obstruction is suspected.
- PSA is a very sensitive test for prostate conditions such as BPH and prostatitis and correlates most often with prostate volume.
- Urine cytology is not recommended during initial evaluation and screening for AMH; it is quite specific for high-grade urothelial carcinoma.
- AFP, β hCG and LDH should be obtained whenever testicular malignancy is suspected.
- A 24-hour urine collection is indicated in high-risk kidney stone formers.
- The functional status of all adrenal masses should be biochemically assessed for hypercortisolism, hyperaldosteronism and pheochromocytoma.

Office Diagnostic Procedures**Uroflowmetry and Ultrasound for Postvoid Residual**

- Uroflowmetry (standing or seated): **maximum flow rate (Qmax)**, mean flow, voided volume, curve — assesses **bladder outlet obstruction (BOO)** in suspected BPH.

QMAX INTERPRETATION

Qmax (mL/s)	Interpretation
>20	Low probability of BOO
15–20	Low probability; significant symptoms may still warrant work-up
10–15	Equivocal
<10	Often BOO, urethral stricture or detrusor impairment

- **Interpret low flows with caution if voided volume is <100 mL.**
- Postvoid residual (PVR): **handheld bladder scanner**; otherwise sterile catheter (invasive — discomfort, trauma, infection). Thresholds poorly defined, but **<100 mL generally acceptable**, especially in older men.
- High PVR: BOO, BPH, detrusor hypocontractility, neurogenic bladder, bladder diverticula, urethral stricture. Further testing follows the complete examination.

Concept — *Free flow reflects both detrusor contraction and outlet resistance, so a low Qmax cannot by itself separate obstruction from detrusor impairment — that is the job of pressure–flow studies.*

Cystometrography and Multichannel Urodynamic Studies

- Urodynamics = study of **transport, storage and evacuation** of urine; indicated when storage or voiding dysfunction is **not diagnosed by history and examination**.
- Performed in the office; ambulatory studies in selected cases. Components: **cystometrography, electromyography, urethral pressure profile, pressure–flow studies**.

Cystourethroscopy

- **Flexible cystoscopy is better tolerated**; rigid optional in women. Diagnostic, screening, some treatments.
- Commonest indications: **lower urinary tract malignancy, urinary obstruction**, assisting urethral catheter insertion.
- **Looposcopy** of an ileal conduit with a flexible cystoscope; generally well tolerated, mild discomfort with significant pressure and reflux into the upper tracts.
- **Antimicrobial prophylaxis is no longer recommended for simple diagnostic cystourethroscopy without manipulation** in patients without infectious risk factors (Table 3.3).

TABLE 3.3 RISK FACTORS ASSOCIATED WITH INCREASED RISK FOR INFECTION FOLLOWING UROLOGIC PROCEDURE

Domain	Risk factors
Host	Advanced age; poor nutritional status; smoking; chronic corticosteroid use; immunodeficiency
Urinary tract / devices	Anatomical anomalies of the urinary tract; externalised catheters; colonised endogenous or exogenous material
Systemic / hospital	Distant coexistent infection; prolonged hospitalisation

Lightner et al., 2020.

- **Purpose:** defines who still needs prophylaxis for simple office procedures. **Trap:** prophylaxis is not routine for every cystoscopy — only in the presence of these factors.

KEY POINTS

- Uroflowmetry and assessment of PVR should be ordered when lower urinary tract obstruction is suspected.
- Urodynamic studies provide information on disorders of storage and voiding.
- Routine antimicrobial prophylaxis is not recommended for routine office cystourethroscopy, urodynamics or cystography in the patient without risk factors.

Radiologic Imaging

- Basic imaging in the office; **computed tomography (CT), magnetic resonance imaging (MRI) and nuclear imaging** at a radiology facility.

Ultrasonography

- **Renal ultrasonography:** size, shape, **hydronephrosis**, cysts and masses, echogenicity, vasculature; increasingly performed by urologists in the office.
- **Cortical echogenicity greater than liver or spleen → medical renal disease.**
- Stone history or flank pain → ultrasonography for obstruction, weighing CT detail against convenience.
- **Renal ultrasonography alone is not sufficient for the haematuria work-up** — cross-sectional imaging is needed.
- Bladder ultrasonography rarely ordered diagnostically; assesses large masses, large stones and volume, with limited information.

- **Ureteric ultrasonography is very difficult — do not use it to diagnose ureteric stones or malignancy;** it may detect severe hydroureter.
- Prostate ultrasonography: mostly **with biopsy**; increasingly **with MRI — software fusion in real time or cognitive fusion** — for lesion targeting; also volume in BPH/obstruction.
- Seminal vesicles and ejaculatory duct: **full seminal vesicles with a dilated ejaculatory duct** inform further testing or treatment in ejaculatory and fertility disorders.

Computed Tomography and Magnetic Resonance Imaging

- Cross-sectional abdominopelvic imaging, with or without contrast, for suspected **upper tract and bladder** conditions.
- **Haematuria (microscopic or gross): contrast CT, or MRI if iodinated contrast is significantly contraindicated.**
- **Non-contrast CT is the most sensitive test and gold standard for urinary stones;** stone history + flank pain → non-contrast CT abdomen and pelvis.
- **MRI is not useful for diagnosis or follow-up of stones;** it excels in soft tissue — renal malignancy, **tumour thrombus**, adrenal pathology, cystic lesions, other abdominal, pelvic and retroperitoneal soft-tissue disease.

Intravenous Pyelogram and Plain Radiographs

- Intravenous pyelogram (IVP) and plain films are largely eclipsed by CT (including low-dose protocols) and MRI.
- **IVP when dynamic imaging of upper tract excretion and obstruction is needed and contrast CT is unavailable.**
- **Plain radiograph of kidneys, ureters and bladder (KUB):** initial evaluation and follow-up of **non-emergent** ureteric or renal stones; **more than 70% of stones are radio-opaque.**
- Sequence: non-emergent ureteric stone → initial KUB → **stone visible → follow up with KUB rather than CT.** Non-obstructing renal stones on surveillance can be followed by KUB; symptoms decide when more complex imaging is needed.

KEY POINTS

- Renal ultrasonography can provide basic screening information on hydronephrosis and medical renal disease but is not an adequate stand-alone study for the work-up of haematuria.
- CT without contrast of the abdomen and pelvis is the gold standard for detecting urinary stones.
- A KUB is a useful and easy test for the follow-up of existing non-emergent radio-opaque urinary stones.

Concept — *Imaging is chosen by the question, not the organ: calcium is best seen by CT and followed by plain film, soft tissue by MRI, and ultrasonography screens but never completes a haematuria work-up.*

SUMMARY

- Laboratory testing is critical to the initial evaluation and subsequent management of urological patients; **understanding when to order each test enables efficient care.**
- **History + examination + a properly collected urinalysis** build the differential diagnosis, direct further laboratory tests, office procedures and imaging, and guide treatment.